Classically Trainable Conditional Generation with Product-State IQP Circuits

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Abstract

Instantaneous quantum polynomial-time (IQP) Born machines admit classical training objectives based on efficiently estimated Pauli observables, but the standard construction is unconditional. We condition one shared IQP circuit by replacing its fixed input with a real product state whose rotation angles encode an external class token. We derive an unbiased Monte Carlo estimator for the resulting Pauli-Z expectations, bound its second moment, and show that its sampled objective supports ordinary backpropagation. An unbiased Conditional MMD objective trains 19-qubit models to represent two discrete Gaussian laws over Hamming weight. Across ten matched seeds, angle conditioning separates the two target laws and outperforms both controls after averaging over conditions, but it fails our stricter condition-wise criterion because the basis control specializes to one target. Exact bitstring evaluation also exposes substantial residual within-shell bias. The results establish a focused route to conditional, classically trained IQP generators without claiming hard sampling or quantum advantage for the tested circuits.

1 Introduction

Quantum circuit Born machines model data by measuring a parameterized quantum state, making distribution learning a natural application of variational quantum algorithms [1]. IQP circuits are a particularly structured family: their commuting interactions are simple to describe [2, 3], while sampling hardness is supported only for appropriate asymptotic families and complexity assumptions [4, 5]. This separation has motivated a "train classically, deploy on quantum" program in which Pauli observables are estimated classically during optimization, while sampling is delegated to a quantum circuit [6, 7].

The standard IQP generative construction learns one output law from one fixed input state. Conditional generation instead requires a single parameterized model to represent a family $q_\theta(x \mid c)$ indexed by an external condition c. Giving each condition independent parameters would fit separate generators rather than test conditional control. The technical question is therefore whether the condition can modify the circuit while preserving the probabilistic estimator that makes IQP training attractive. This question also intersects broader limits on gradient measurement, expressivity, and trainability [8–10].

We encode c in a real product state $\bigotimes_i R_Y(\alpha_{c,i})|0\rangle$ and apply one shared HD_ZH IQP circuit (Fig. 1). Product states remain computationally tractable after the first Hadamards: their measurement law factorizes and their amplitudes are products of local terms. Adapting the sampling argument of Van den Nest [11] yields an unbiased amplitude-ratio estimator with bounded second moment. Because its random reference strings do not depend on θ, JAX differentiates the sampled MMD objective directly. We use a small-variance initialization and low-body biased multiscale kernels in light of recent IQP trainability analyses [12, 13].

Our contributions are threefold. First, we state and verify the product-state Pauli estimator and its backpropagated gradient. Second, we construct an unbiased, class-balanced Conditional MMD loss and validate that it detects label swaps invisible to pooled MMD. Third, we evaluate 19-qubit heavy-hex, lattice, and complete two-local ansatz families on two condition-indexed Gaussian Hamming-weight laws, using exact bitstring metrics and ten matched seeds. The topology study compares native families rather than fixed parameter counts; connectivity is operationally relevant to IQP implementations [14], but no hardware or noise claim is made here.

1.1 Scope relative to prior IQP training

The classical simulation result behind this work estimates local observables, not samples from the full output distribution [11]. IQPopt turns that distinction into an optimization strategy by expressing distributional kernels through Pauli characters [6]. We retain this observable interface but change the reference state. The relevant extension is not a new differentiation rule: once reference strings have been sampled, the estimator is a smooth elementary function of θ, so the complete MMD estimate can be differentiated by ordinary reverse-mode automatic differentiation. The work is therefore about preserving an existing classical-training mechanism under conditional product-state inputs.

The condition is deliberately excluded from D_θ. Appending condition-specific diagonal gates would make it unclear whether the experiment learned a shared conditional mechanism or two parameter sets. Here one parameter vector must satisfy both target laws. Independent models are reported only as reference fits, while input states with no distinguishing condition and with computational-basis conditions serve as controlled ablations.

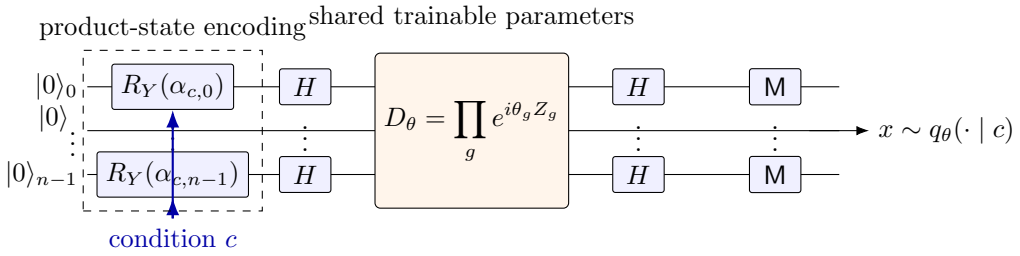


Figure 1: The condition changes only the initial product-state angles. The commuting diagonal block and all trainable parameters are shared across conditions. The experiment broadcasts one scalar angle over all qubits.

2 Conditional product-state IQP model

For n qubits, let $g \in \{0,1\}^n$ denote a Pauli mask, $Z_g = \prod_i Z_i^{g_i}$, and define

$$D_\theta = \prod_{j=1}^{m} e^{i\theta_j Z_{g_j}}, \qquad U_\theta = H^{\otimes n} D_\theta H^{\otimes n}. \tag{1}$$

The condition is represented by $|\psi_c\rangle = \bigotimes_i R_Y(\alpha_{c,i})|0\rangle$ and the model law is $q_\theta(x \mid c) = |\langle x|U_\theta|\psi_c\rangle|^2$. Only $\boldsymbol{\alpha}_c$ depends on c; θ and the generator set are shared. Our general estimator allows arbitrary real angle vectors, while the experiment uses $\alpha_A = 0$ and $\alpha_B = \pi/4$ broadcast over all qubits.

2.1 Reference-state sampling

Write $|\phi_c\rangle = H^{\otimes n}|\psi_c\rangle$. Its local amplitudes are

$$\phi_{c,i}(0) = \frac{\cos(\alpha_{c,i}/2) + \sin(\alpha_{c,i}/2)}{\sqrt{2}}, \tag{2}$$

$$\phi_{c,i}(1) = \frac{\cos(\alpha_{c,i}/2) - \sin(\alpha_{c,i}/2)}{\sqrt{2}}. \tag{3}$$

Consequently $\phi_c(z) = \prod_i \phi_{c,i}(z_i)$, so both sampling $z \sim |\phi_c(z)|^2$ and evaluating amplitudes take $O(n)$ local operations. This is the sole tractability property required from the encoding.

3 Classical estimation and Conditional MMD

3.1 Pauli expectation estimator

For an observable mask $a \in \{0,1\}^n$, define the phase $\Phi_\theta(z) = \sum_j \theta_j (-1)^{g_j \cdot z}$. Sampling $z \sim |\phi_c(z)|^2$, set

$$F_{a,c}(z) = \frac{\phi_c(z \oplus a)^*}{\phi_c(z)^*} e^{i[\Phi_\theta(z) - \Phi_\theta(z \oplus a)]}, \tag{4}$$

with zero-probability denominator events omitted.

Proposition 1. *The real part of the sample mean of Eq. (4) is an unbiased estimator of $\langle\psi_c|U_\theta^\dagger Z_a U_\theta|\psi_c\rangle$. Moreover $\mathbb{E}|F_{a,c}|^2 \le 1$, hence $S = O(\epsilon^{-2})$ samples suffice for constant-probability additive error ϵ. Sampling and evaluating each contribution is polynomial in n and the number of generators.*

Proof. Multiplying Eq. (4) by $|\phi_c(z)|^2$ cancels one amplitude ratio and summing over z gives the computational-basis expansion of $\langle Z_a\rangle$. For the second moment,

$$\mathbb{E}|F_{a,c}|^2 = \sum_{z:\phi_c(z)\neq 0} |\phi_c(z \oplus a)|^2 \le 1. \tag{5}$$

The sample-complexity statement follows from the variance of a sample mean. Product structure gives the stated per-sample computational cost. □

Ordinary backpropagation applies because the reference samples are independent of θ. Differentiation only multiplies a contribution by $i[\chi_{g_j}(z) - \chi_{g_j}(z \oplus a)]$, where $\chi_g(z) = (-1)^{g \cdot z}$; its magnitude is at most two. Thus automatic differentiation of the Monte Carlo estimate yields an unbiased gradient estimate without parameter-shift circuit evaluations.

3.2 Unbiased Conditional MMD

For binary strings we use $k_\sigma(x,y) = \exp[-d_H(x,y)/(2\sigma^2)]$. Its Fourier representation samples each bit of a independently with probability

$$p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{6}$$

and writes MMD as an expectation of squared differences between target and model Pauli characters. We average $\sigma \in \{1,2,4\}$ rather than select one data-dependent bandwidth [15]. For target samples grouped by condition, the training objective is

$$\mathcal{L}(\theta) = \frac{1}{2} \sum_{c \in \{A,B\}} \mathbb{E}_{\sigma,a}\left[\left(\langle Z_a\rangle_{p_c} - \langle Z_a\rangle_{q_\theta(\cdot|c)}\right)^2\right]. \tag{7}$$

We use U-statistics for both squared terms, making the finite target-sample and model Monte Carlo estimate unbiased. Negative stochastic values are therefore possible and are neither clipped nor square-rooted.

3.3 Why direct backpropagation is valid

Let $\widehat{\mathcal{L}}(\theta;\omega)$ denote one stochastic loss, where ω contains target minibatches, Fourier masks, and product-reference samples. Their sampling laws are independent of θ. The estimator and its derivative are bounded for the finite generator set, so differentiation and expectation may be interchanged:

$$\nabla_\theta \mathcal{L}(\theta) = \nabla_\theta \mathbb{E}_\omega[\widehat{\mathcal{L}}(\theta;\omega)] = \mathbb{E}_\omega[\nabla_\theta \widehat{\mathcal{L}}(\theta;\omega)]. \tag{8}$$

Thus we backpropagate through the sampled MMD exactly as in regular observable-trained IQP models. "Unbiased" refers to expectation over ω; a single stochastic loss can be negative because its two squared terms are U-statistics. Taking a square root or clipping at zero would destroy this property and introduce a non-differentiable boundary.

3.4 Finite-sample construction

For completeness, fix one condition, bandwidth, and Fourier mask. Given target strings $x_1, \dots, x_R$, define $t_r = (-1)^{a \cdot x_r}$. Given independent product-reference samples $z_1, \dots, z_S$, define $f_s = \operatorname{Re} F_{a,c}(z_s)$. We estimate the squared character discrepancy with

$$\widehat{\Delta}^2_{a,c} = \frac{1}{R(R-1)} \sum_{r \neq r'} t_r t_{r'} + \frac{1}{S(S-1)} \sum_{s \neq s'} f_s f_{s'} - \frac{2}{RS} \sum_{r,s} t_r f_s. \tag{9}$$

The three terms estimate the target square, model square, and cross product without diagonal self-products. Independence therefore gives $\mathbb{E}[\widehat{\Delta}^2_{a,c}] = (\langle Z_a \rangle_{p_c} - \langle Z_a \rangle_{q_c})^2$. The implementation averages independent instances over masks, bandwidths, and conditions. This explicit construction is also why a negative minibatch loss is not evidence of an invalid kernel: an unbiased estimate of a nonnegative quantity need not itself be nonnegative.

Product-state sampling is vectorized over z, while the character table $(-1)^{g_j \cdot z}$ evaluates all commuting generators together. For each parameter, the derivative multiplier has magnitude at most two; combined with Proposition 1, its per-sample second moment is at most four. The chosen experimental angles keep every local reference amplitude nonzero. The estimator still covers limiting angles with zero amplitudes because such denominator events have zero sampling probability and the second-moment proof explicitly restricts to the support.

4 Experiments

4.1 Targets, ansätze, and protocol

The two targets are distributions over 19-bit strings. Their total mass on weight k is proportional to $\exp[-(k - \mu'_c)^2/(2\sigma_t^2)]$, with $\sigma_t = \sqrt{19}/2$; mass is uniform among strings of the same weight. We numerically calibrate μ'_c after truncation to $k \in [0, 19]$ so the exact means are $\mathbb{E}[K|A] = 19/2$ and $\mathbb{E}[K|B] = 19/4$. Each condition supplies 10,000 training and 2,000 disjoint validation strings.

More explicitly, if $\rho_c(k)$ is the normalized shell mass, then every string x receives

$$p_c(x) = \frac{\rho_c(|x|)}{\binom{19}{|x|}}. \tag{10}$$

Thus the task requires a Gaussian-shaped binary distribution over Hamming weight and exact uniformity within each shell. The location μ'_c is found by one-dimensional root solving after truncation; σ_t is the specified Gaussian width, while reported standard deviations are recomputed from the normalized discrete target itself. Samples are produced by drawing a shell from ρ_c and then choosing a uniformly random subset of that size.

Every ansatz includes one independent Z_i angle and one independent $Z_i Z_j$ angle per interaction edge. The distance-3 heavy-hex, corner-trimmed 4×5 lattice, and complete graphs have 39, 48, and 190 parameters, respectively. Adam runs for 2,000 steps with cosine decay from 10^{-2} to 10^{-3}. Parameters are initialized as $\mathcal{N}(0, 0.1^2/m)$, and ten matched seeds share data, minibatches, observable masks, and Monte Carlo keys across topologies. Calibration selected 64 observables and 8,192 reference samples per bandwidth for training; exact small-system tests gave expectation RMSE 0.0085.

4.2 Baselines, evaluation, and decision rule

All conditioning ablations use the 190-parameter complete graph. The no-condition control feeds $\alpha = \pi/8$ for both labels, so one output law must compromise between the targets. The basis control feeds $|0\rangle^{\otimes n}$ for A and $|1\rangle^{\otimes n}$ for B. Two independent complete-graph models, trained only on A or only on B with the corresponding angle encodings, provide reference fits but are not conditional baselines.

The basis control has an important exact restriction. Since $H^{\otimes n} X^{\otimes n} = Z^{\otimes n} H^{\otimes n}$ and D_θ commutes with $Z^{\otimes n}$, its two output laws are bitwise complements:

$$q_\theta(x \mid B) = q_\theta(x \oplus 1^n \mid A). \tag{11}$$

It can therefore specialize to the symmetric target centered at $n/2$, but then its other Hamming-weight mean is also $n/2$ rather than $n/4$. This baseline tests whether continuous angle encoding does more than a discrete input flip; it is not expected to be uniformly weak.

For a target p and exact model law q, bitstring TVD is $\frac{1}{2} \sum_x |p(x) - q(x)|$. Shell TVD applies the same definition after aggregating by $|x|$, and mean and standard-deviation errors compare the first two moments of that shell law. Exact model probabilities are obtained for all 2^{19} strings by applying a fast Walsh–Hadamard transform to $e^{i\Phi_\theta(z)} \phi_c(z)$; no Monte Carlo estimate enters these test metrics. Held-out Conditional MMD is reported only as an objective-space diagnostic.

Before running the matrix, we fixed a strict success rule: for each condition separately, the complete angle-conditioned model must have lower median shell TVD than both conditioning controls, and bitstring TVD must show the same ordering. Results are summarized by median and interquartile range over ten seeds. A balanced average over conditions is informative but cannot override this condition-wise decision rule.

4.3 Results

Table 1 reports medians and interquartile ranges over ten seeds. The primary comparison uses exact enumera-

tion of all 2^{19} outputs. Shell TVD measures the stated Gaussian-weight task; bitstring TVD additionally tests uniformity inside every shell.

Conditioning test. The strict criterion fails. For condition A, complete angle conditioning reaches shell TVD 0.121, compared with 0.403 for no conditioning but 0.007 for the basis control. Its bitstring TVD is nevertheless lower than both controls (0.352 versus 0.497 and 0.405). For condition B, angle conditioning wins on shell TVD (0.072 versus 0.327 and 0.722) and bitstring TVD (0.275 versus 0.462 and 0.778). All four required comparisons except the A shell comparison pass. Averaged over conditions, the angle model has median shell and bitstring TVDs of about 0.096 and 0.314, substantially below either control, but these balanced summaries are descriptive rather than a post-hoc replacement for the criterion.

Topology and sharing. Topology rankings depend on the resolution of evaluation. The lattice has the best balanced shell TVD (about 0.085), narrowly ahead of heavy-hex (0.088) and complete (0.096), whereas the complete graph has the best balanced bitstring TVD (0.314 versus 0.397 and 0.445). Since the families contain 48, 39, and 190 parameters, respectively, this is not a connectivity-at-fixed-capacity experiment. Independent reference models obtain shell TVDs 0.018 on A and 0.036 on B, showing a measurable cost to using one shared parameter vector. Their presence also guards against attributing every residual error to the estimator or optimizer.

Distributional diagnosis. The learned weight laws are substantially closer than the full bitstring laws: correct Hamming-weight histograms can coexist with nonuniform probability inside a shell. This is scientifically important because the target definition requires both. The independent edge parameters do not enforce permutation symmetry, and the low-body-biased MMD kernels emphasize coarse moments; the result is therefore a conditional proof of concept rather than complete distribution recovery. The complete angle model matches the target means closely (absolute errors 0.041 and 0.036), while its standard-deviation errors remain 0.598 and 0.181 for A and B, respectively.

4.4 Estimator checks and reproducibility

We checked the estimator against exact six-qubit statevectors and verified automatic-differentiation gradients against finite differences. A separate brute-force small-system test forms the kernel U-statistic directly and compares both its stochastic value and backpropagated gradient with the Fourier-observable implementation. Twenty-repeat calibration selected the smallest tested expectation budget below RMSE 0.01 and the smallest observable budget below 5% MMD coefficient of variation. Training, validation, and exact evaluation are separate code paths; the final distribution metrics are recomputed from each best validation checkpoint.

Matched randomness reduces variance in topology comparisons without making the ten runs identical: parameters are initialized independently for each seed, while data generation and stochastic estimator keys are shared across the models being compared. All target calibration, training, exact enumeration, aggregation, criterion auditing, and figure-generation scripts are included with machine-readable per-seed outputs.

5 Discussion and conclusion

Product-state conditioning extends the observable-training mechanism of IQP Born machines without introducing condition-specific trainable parameters. The estimator is classically efficient for the stated Pauli objective, and the experiment tests conditional control with exact distributional metrics. It does not meet the preregistered condition-wise success rule. The failure is diagnostic rather than ambiguous: the basis control's complement symmetry is exceptionally well matched to A and fundamentally mismatched to B, while angle conditioning provides the much better balanced model.

Three limitations shape the result. First, the topology comparison does not isolate connectivity from capacity: its families have different parameter counts. Second, the target is permutation symmetric but the parameterization is not, which helps explain the gap between shell and bitstring TVD. Third, hardware routing and noise are not simulated, and exact 2^{19} enumeration would not remain an evaluation strategy at large n.

5.1 What the failure reveals

The reference distribution seen by the diagonal circuit makes the effect of the encoding concrete:

$$\Pr_{\phi_c}(z_i = 1) = |\phi_{c,i}(1)|^2 = \frac{1 - \sin \alpha_{c,i}}{2}. \qquad (12)$$

Condition A uses $\alpha_A = 0$ and hence the uniform IQP reference law, whereas B uses $\alpha_B = \pi/4$ and a biased product reference law. The primary and basis models therefore use exactly the same input for A; they differ in the second input and in the compromises imposed on their shared parameters. This explains why the basis result on A is a serious control, not a comparison to an unrelated architecture. Its near-zero shell error says that the complete ansatz and optimizer can fit the central Gaussian when the second condition supplies no compatible requirement. Its B failure follows from the exact complement constraint. The angle model gives up some A shell accuracy to move B from the complementary mean $n/2$ toward $n/4$.

The multiscale kernel also predicts which discrepancies will be easiest to learn. At $\sigma = 1, 2, 4$, a Fourier mask includes each bit with probabilities approximately 0.197, 0.059, and 0.015, corresponding at $n = 19$ to expected mask weights 3.74, 1.12, and 0.29. Training therefore emphasizes low-order characters. Those observables efficiently constrain coarse mass and moments

Table 1: Exact test metrics, median [interquartile range] over ten seeds. Lower is better. Values are inserted from the reproducible result summary.

Model	Condition	Parameters	Bitstring TVD	Shell TVD	Mean error	Std. error
Angle / heavy-hex	A	39	0.487 [0.476,0.500]	0.066 [0.065,0.073]	0.21 [0.17,0.23]	0.26 [0.24,0.28]
Angle / heavy-hex	B	39	0.402 [0.398,0.409]	0.109 [0.107,0.114]	0.37 [0.30,0.50]	0.32 [0.28,0.35]
Angle / lattice	A	48	0.383 [0.350,0.409]	0.064 [0.055,0.073]	0.15 [0.14,0.18]	0.24 [0.21,0.30]
Angle / lattice	B	48	0.411 [0.406,0.419]	0.106 [0.104,0.109]	0.24 [0.18,0.29]	0.38 [0.36,0.38]
Angle / complete	A	190	0.352 [0.339,0.356]	0.121 [0.112,0.126]	0.04 [0.03,0.08]	0.60 [0.54,0.61]
Angle / complete	B	190	0.275 [0.271,0.283]	0.072 [0.067,0.074]	0.04 [0.02,0.06]	0.18 [0.17,0.21]
No condition	A	190	0.497 [0.494,0.512]	0.403 [0.400,0.407]	2.41 [2.38,2.43]	1.06 [1.01,1.08]
No condition	B	190	0.462 [0.460,0.465]	0.327 [0.324,0.331]	2.34 [2.32,2.37]	1.12 [1.07,1.15]
Basis condition	A	190	0.405 [0.374,0.427]	0.007 [0.006,0.008]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
Basis condition	B	190	0.778 [0.771,0.783]	0.722 [0.722,0.723]	4.75 [4.75,4.75]	0.06 [0.06,0.06]
Separate A	A	190	0.363 [0.359,0.393]	0.018 [0.016,0.019]	0.08 [0.07,0.09]	0.00 [0.00,0.00]
Separate B	B	190	0.195 [0.182,0.222]	0.036 [0.031,0.041]	0.04 [0.02,0.06]	0.04 [0.03,0.06]

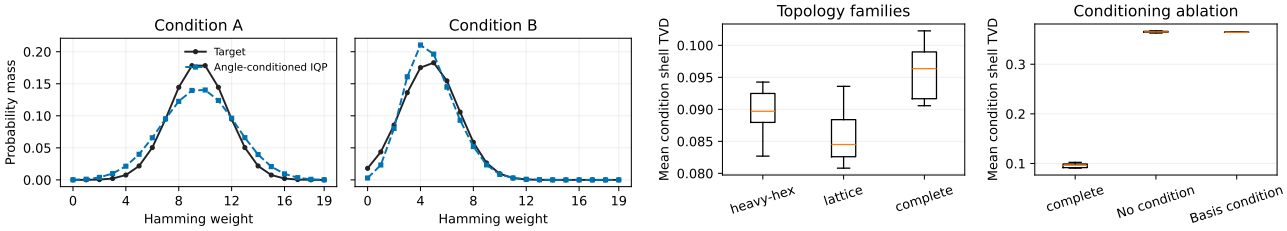


Figure 2: Left: target and representative complete-ansatz weight laws. Right: ten-seed topology and conditioning comparisons.

but do not force equality among the $\binom{19}{k}$ strings within each shell. The exact TVD gap is consistent with this spectral bias. It should not be read as a defect hidden by Monte Carlo: the gap appears in complete enumeration after training.

5.2 Failure-directed next experiments

The most direct follow-up is to align the parameterization with the target symmetry. Tying all one-qubit generators and tying two-qubit generators by a graph automorphism orbit would remove arbitrary within-shell preferences and sharply reduce parameter count. A more flexible alternative is to randomize qubit permutations during training, retaining heterogeneous parameters while making asymmetry costly. Either intervention should be tested against the same bitstring TVD, since shell-only evaluation cannot establish the desired uniformity.

A second experiment should capacity-match the topology families, for example by adding repeated sparse layers until their parameter counts are comparable. The present reversal—sparse graphs lead in shell TVD while the complete graph leads in bitstring TVD—could arise from connectivity, capacity, or optimization geometry. The current native-family study intentionally does not separate them.

Finally, conditional generalization remains open. The estimator accepts arbitrary per-qubit real angles, but this experiment trains only two broadcast prototypes. Training on several angles and withholding intermediate values would test whether product-state geometry supports interpolation rather than two-point memorization. At larger n, exact TVD must give way to symmetry-aware shell tests, held-out characters of increasing weight, and carefully qualified sample-based two-sample tests; none alone is a replacement for the exact bitstring audit available here.

Efficient training must also be distinguished from efficient classical sampling. Noise, anticoncentration, and verification create delicate trade-offs for IQP generative models [4, 13]; our 19-qubit exact evaluation provides no hardness evidence. Natural next steps are permutation-tied generators, richer condition vectors, and larger experiments whose evaluation avoids complete enumeration. Within its stated boundary, this work shows that changing the tractable initial state is enough to turn one shared IQP circuit into a classically trainable conditional generator.

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